

SENEGAL
COUNTRY REPORT ON GENOME EDITING LANDSCAPE
ANALYSIS ON AGRICULTURE

MAY 2026

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LIST OF ABBREVIATIONS/ACRONYMS

AATF	African Agricultural Technology Foundation
ABS	Access and Benefit Sharing
ANAQ-Sup	National Authority for Quality Assurance in Higher Education
ASPIT	Senegalese Agency for Industrial Property and Technological Innovation
AUDA-NEPAD	African Union Development Agency- New Partnership for Africa's Development
BSL	Biosafety Level
BT	Breeding Technique
CBD	Convention on Biological Diversity
CGIAR	Consultative Group on International Agricultural Research
CIGASS	Centre International de Recherche et de Formation en Génomique Appliquée et de Surveillance Sanitaire
CILSS	The Permanent Inter-State Committee for Drought Control in the Sahel
CIRAD	French Agricultural Research Center for International Development
CNERS	National Ethical Committee for Health Research
CNRA	National Center for Agronomic Research
COOPROFEL	Cooperative of Fruit, Vegetables and Livestock Producers
CSAIP	Climate Smart Agriculture Investment Plan
CSE	Ecological Monitoring Centre (CSE)
ECOWAS	Economic Community of West African States
FAO	Food and Agriculture Organization of the United Nations
GE	Genetically Engineered
GE _d	Gene Editing
GIZ	German Corporation for International Cooperation
GMO	Genetically Modified Organism
IAEA	International Atomic Energy Agency
IP	Intellectual Property
IPRs	Intellectual Property Rights
IRD	Institut de Recherche pour le Développement
IRESSEF	Institut de Recherche en Santé de Surveillance Épidémiologique et de Formation
ISRA	Senegalese Institute of Agricultural Research
ITA	The Institute of Food Technology
MEET	Ministry of Environment and Ecological Transition
MESRI	Ministry of Higher Education, Research and Innovation
MoA	Ministry of Agriculture

NBA	National Biosafety Authority
NBC	National Biosafety Committee
OAPI	African Intellectual Property Organization
ODK	Open Data Collection Kit
PSE	Emerging Senegal Plan
RESOPP	Network of Peasant and Pastoral Organizations of Senegal
R&D	Research and Development
SAF	Sustainable Agriculture Foundation (formerly Syngenta Foundation)
SEDAB	Sahélienne d'Entreprise de Distribution et d'Agro-business
SSA	Food Sovereignty Strategy
SSG	Seed Systems Group
UCAD	Université Cheikh Anta Diop de Dakar (Cheikh Anta Diop University)
UNFCCC	United Nations Framework Convention on Climate Change
UNIS	National Inter-professional Seed Union
USAID	United States Agency for International Development

EXECUTIVE SUMMARY

Background: Agriculture is vital to Senegal, serving as a cornerstone of the economy by employing 60% of the population, including 752,352 family farms, and contributing significantly to GDP. Senegal's key national agricultural plan is the Climate Smart Agriculture Investment Plan (CSAIP), a collaborative effort to transform agriculture by integrating climate resilience, sustainable practices, and enhanced productivity. Due to droughts driven by climate change, national production does not meet population needs. In 2022, Senegal enacted a new biosafety law that repealed the 2009 Biosecurity Law and established a process for the entry, research, and commercialization of genetically engineered products, including genome-edited (GE_d) products. For a long time, Senegal relied on conventional breeding techniques to develop crop varieties to address productivity challenges, with limited success. As a result, there is a growing need to adopt new breeding techniques, such as GE_d, to address challenges in crop productivity.

The African Union Development Agency – New Partnership for Africa Development (AUDA-NEPAD) commissioned an Assignment on the Genome Editing Landscape Analysis of Senegal to provide an in-depth assessment of existing policies, infrastructure, institutions, and technical capabilities to support product development and commercialization. The specific objectives of the assignment were to: 1) Provide an evidence-based description and analysis of the status of modern biotechnology and GE_d in Senegal, highlighting key trends, intervening factors, and areas for attention, as well as fundamental aspects such as science/technical, political, geopolitical, social, human, cultural, and traditional factors that support or hinder advances in the application of GE_d in agriculture and food systems; 2) Identify the emerging needs (i.e., agricultural productivity, reduction of postharvest losses, climate adaptation, food and nutrition security, diversified and healthy diets) in Senegal that GE_d can readily address, especially those that require rapid responses at scale; and 3) Identify staple and indigenous crops based on the Senegal national context that can improve the livelihoods of people through food security, better nutrition, climate resilience, and sustainable productivity.

Data were collected from both secondary and primary sources, including published literature, stakeholders' (institutional) databases, and websites. Primary data were collected through live interviews using an Online Data Collection Kit (ODK), as well as surveys and questionnaires. The generated data were analyzed, synthesized, and packaged in terms of signatory to multilateral food and environmental agreements/treaties, regulatory frameworks, socio-economic considerations in decision-making, GE_d projects and programs, and existing human and institutional capacity. Additionally, data were categorized into research and development, academic institutions, crops for improvement through GE_d, intellectual property rights, and funding sources.

Key findings: The GEd Landscape Analysis revealed that Senegal is a signatory to the Codex Alimentarius Commission (CAC), the United Nations Framework Convention on Climate Change (UNFCCC), the Convention on Biological Diversity (CBD), the Cartagena Protocol on Biosafety (CPB), the Nagoya-Kuala Lumpur Supplementary Protocol on Liability and Redress, and the Nagoya Protocol on Access and Benefit Sharing. Senegal has domesticated these food and environmental management frameworks to demonstrate its commitment to global efforts in food safety, the environment, and biodiversity conservation.

Socioeconomic issues in decision-making are being addressed by establishing the national biosecurity committee. The socioeconomic considerations focus on food and nutritional security, the development of new crop varieties, the development of climate-resilient animal breeds, and biodiversity conservation. The survey found that Senegal currently has no modern biotechnology or GEd projects in the research and development (R&D) or commercialization stages.

Regarding human capacity development, two scientists from Cheikh Anta Diop University and ISRA were trained at IRD (Institut de Recherche pour le Développement) in Montpellier to support the initiation of GEd research in Senegal. Additionally, three scientists received limited training from the Institut de Recherche en Santé de Surveillance Epidémiologique et de Formation (IRESSEF), CIGASS, the Department of Plant Biology's plant biotechnology campus laboratory at Cheikh Anta Diop University (UCAD), and the National Biosafety Authority (NBA). Of the eight universities assessed in Senegal, only Cheikh Anta Diop University had a well-equipped molecular laboratory capable of integrating GEd into its research portfolio.

The study found that ISRA is the only major national agricultural research organization/institute (NARO/NARI) in Senegal. Other R&D institutions in Senegal include AfricaRice, IRD, and CIRAD. These institutions were found to have poorly equipped, under-resourced laboratories (in terms of funding and personnel) and to lack specialized equipment for biotech/GEd-related activities.

Several indigenous and commercial crops were identified as having greater potential for GEd research in Senegal. They include cowpea, pepper, rice, mango, tomato, peanut, and millet. Crop yields could be improved by using GEd technology to enhance tolerance or resistance to biotic and/or abiotic stresses.

Two institutions administer IPRs in Senegal. These include the Agency for Industrial Property and Technological Innovation (ASPIT), which provides patent protection for inventors and innovators, and Trademark Senegal, which regulates trademarks and industrial property rights. Senegal is a member of the African Intellectual Property Organization (OAPI).

A few seed companies play a crucial role in providing high-quality seeds to smallholder farmers in Senegal. These include Tembédou et Fils, SEDAB, Tropicasem, Bayer, and

Syngenta. However, only Bayer and Syngenta advocate the adoption of agricultural biotechnology. The recent approval of 18 decrees to implement the biosafety law paves the way for seed companies to conduct biotech/GE^d R&D to improve the productivity and nutritional quality of key indigenous and commercial crops. Funding for biotechnology-related agricultural research in Senegal comes from various sources, including international organizations, USAID, the International Atomic Energy Agency (IAEA), and the government. Key players in this area include the National Biosafety Authority (NBA) and AUDA-NEPAD, which are working to enhance awareness and build public trust in the ability of Senegalese agencies to regulate modern agricultural biotechnology products for research, production, and trade.

Policy Implications

Senegal passed Biosafety Law No. 2022-20 to modernize its legal framework for genetic technologies. To build on this foundation, the following phased policy options are critical:

Short term (next 12–24 months)

- **Finalize Implementing Decrees:** The government must finalize the draft decrees for Law No. 2022-20 to provide operational clarity.
- **Product-based Regulatory Exemptions:** Create specific biosafety guidelines that exempt gene-edited products from standard GMO regulations if the final product is functionally identical to conventionally bred crops.
- **Streamline Lab Procurement:** Simplify customs procedures and reduce duties or taxes on laboratory reagents and equipment to accelerate research.

Medium to long term (2–5 years)

- **Regional Expertise Pooling:** Collaborate with organizations like AUDA-NEPAD and CGIAR to pool regional expertise, fostering large-scale research programs that tackle climate resilience and food security.
- **Public Engagement and Consumer Confidence:** Implement transparent, multi-stakeholder public awareness campaigns and clear labelling frameworks to address socio-political concerns and foster societal acceptability regarding the benefits of the modern biotechnology and GE^d.

Conclusion

Biotech and GE^d can offer Senegal many opportunities to enhance food security, improve healthcare, drive economic growth, and address environmental challenges. Biotech/GE^d

could become a transformative force for sustainable development in Senegal as the country grapples with rapid population growth, climate change, and endemic crop and animal diseases. Senegal has a regulatory framework for handling and regulating GMO and GEd products. However, the law's effectiveness depends on the approval of 18 executive decrees for its implementation. A process has begun to consolidate the executive decrees to expedite their signature by the president of Senegal. As a result, no biotech/GEd activities can be carried out in the country.

1. INTRODUCTION

1.1. Description of Senegal's Agricultural Landscape

Agriculture is the primary employer, accounting for 60% of the workforce and including 752,352 family farms. In 2013, 75% of farms were between 1 and 5 hectares, while 5.5% were less than 1 hectare. Two types of crops are grown: cash crops (horticulture, peanuts, cotton) and food crops (millet, sorghum, corn, rice). At the macroeconomic level, the primary agricultural sector accounted for approximately 16.37% of GDP in 2023 (O'Neill, 2025). However, the country's agricultural sector relies heavily on rainfall. Droughts driven by climate change mean national production does not meet population needs. To address these challenges, classical breeding techniques were used to develop new crop varieties that increased national production. However, these varieties did not respond effectively to changing climatic conditions, underscoring the need for new breeding techniques such as GEd.

1.2 National Frameworks and Investments

The National Biosafety Authority (NBA), created by Law No. 2009-27 of July 13, 2009 (NBA, 2009), is responsible for regulating the development, contained use, import, export, transit, transport, deliberate release into the environment, and placing on the market of genetically modified organisms or derivatives. The NBA is under the supervision of the Ministry of Environment and Ecological Transition (MEET). The NBA's Executive Director is appointed by presidential decree, upon the MEET's advice. Through the National Biosafety Committee (NBC), the NBA is responsible for risk assessments related to the import, export, handling, transit, confined use, release, or launch of GMOs or derived products. The NBC's 26 members include scientists, representatives of the public and private sectors, and members of the general public. A new committee will replace the current NBC with more scientists specializing in risk assessment. This change is intended to address the rapid evolution of New Breeding Techniques. Senegal's regulatory framework for handling and regulating GMO and GEd products is the Biosafety Law (n° 2022-20 of June 14th, 2022) and the Regulation C/REG. 04/09/2020 related to GMO and biotechnology products in the ECOWAS (Economic

Community of West African States) and CILSS (Inter-State Committee for Drought Control in the Sahel) (FAO, 2020) (see Table 1).

1.3. Engagement in Regional Frameworks

Senegal is a member of the Economic Community of West African States (ECOWAS), which has adopted the basic Regulation C/REG. 04/09/2020 related to GMO and biotechnology products. ECOWAS promotes harmonization of biotechnology regulations and facilitates regional dialogue. Senegal has participated in ECOWAS biosafety consultations and cross-border risk communication workshops. Senegal is active in African Union biotech efforts, particularly the AUDA-NEPAD African Biosafety Network of Expertise (ABNE), which supports capacity-building in biosafety and genome-editing policy. Through WAEMU, ECOWAS, and AU frameworks, Senegal is part of efforts to create enabling environments for responsible genome editing.

1.4. Challenges and Opportunities: Justification of GEd Assessment based on Current Issues Identified

1.4.1 Challenges Justifying GEd Assessment

Despite its benefits, biotech/GEd technology in Senegal is hindered by regulatory frameworks, limited financial investment, human capacity constraints, limited laboratory infrastructure, and ethical concerns. Harmonizing regulatory frameworks with ECOWAS neighbors, investing in scientific education, training, and infrastructure, and fostering regional and international collaboration are critical steps to address these obstacles.

1.4.2 Opportunities Supporting GEd Investment

Biotech and GEd offer Senegal significant opportunities to enhance food security, improve healthcare, drive economic growth, and address environmental challenges. Biotech/GEd could become a transformative force for sustainable development in Senegal as the country grapples with rapid population growth, climate change, and endemic crop and animal diseases.

1.5. Objectives of the Landscape Analysis

The general objective of the GEd Landscape Analysis was to conduct an in-depth assessment of existing policy, infrastructure, institutional, and technical capabilities supporting product development and commercialization across a selected set of African countries.

Specifically, for Senegal, as in the other selected countries, the objectives of the Landscape Analysis were to:

- i. Provide an evidence-based description and analysis of the status of modern biotechnology and GEd in Senegal, highlighting key trends, intervening factors, and fundamental aspects such as science/technology, politics, geopolitics, social and human factors, culture and traditions, etc., that support or hinder advances in the application of genome editing in agriculture and food systems.
- ii. Identify the emerging needs in Senegal that GEd can readily address, particularly those requiring rapid, scale-wide responses. These needs will focus on food systems, including agricultural productivity, reduction of postharvest losses, climate adaptation, food and nutrition security, diversified and healthy diets, and
- iii. Identify staple and indigenous crops in the Senegalese national context that can improve livelihoods through food security, improved nutrition, climate resilience, and sustainable productivity.

2. APPROACH / METHODOLOGY

Data were collected from both secondary and primary sources. Secondary data were gathered from published literature and institutional databases, including official reports, peer-reviewed publications, and relevant organizational websites. Primary data were collected through live interviews using the Online Data Collection Kit (ODK), structured surveys, and email-based questionnaires shared in Microsoft Word format. Primary data collection served to validate, supplement, and strengthen the secondary data. Data from both sources were subsequently analysed, synthesized, and packaged under the following thematic areas:

2.1. Regulatory and Policy Frameworks for Biotechnology and Genome Editing

Components of the biotechnology and genome-editing regulatory and policy frameworks were identified and documented using secondary sources (published literature and institutional websites) and primary data obtained through interviews, surveys, and email correspondence.

2.2. Genome Editing Projects

Information on biotechnology and genome-editing projects, particularly genome-editing initiatives, was compiled from primary and secondary data sources. This included documentation of target crops, livestock, fisheries, and forestry species; traits under development; key stakeholders and partnerships involved; and funding sources. The synthesized data were used to:

- i. identify emerging economic, social, and environmental/climate-related needs; and
- ii. Assess the status of human capacity and infrastructure supporting genome editing technologies in the target countries.

2.3. Traditional and Staple Crops for Genome Editing Improvement

Data on genome editing projects were further disaggregated to identify staple, indigenous, and commercial crops with the highest potential for socio-economic impact. Selection criteria included national acceptance, resource requirements, scalability, and likelihood of successful development and deployment.

2.4. Institutional Capacity (Human Capital, Infrastructure and Equipment)

Data on institutional capacity, including human resources, laboratory and field infrastructure, and equipment for genome editing R&D, commercialization, and scaling, were collected from respondents during the primary data collection exercise.

2.5. Stakeholder Mapping

Key stakeholders and institutions, and, where appropriate, individual experts, were identified to provide critical information on existing biotechnology and genome-editing interventions. Stakeholders were categorized into five (5) groups, as reflected in the data collection tools:

- Regulatory agencies
- Research organizations and institutions.
- Universities
- Private sector and industry
- Government ministries, agencies, and policymakers

2.6. Database Systems and Data Management

Customized data collection tools (questionnaires) and digital platforms were developed to support primary data collection. Questionnaires were tailored to each stakeholder category and used to generate structured datasets assessing Senegal's preparedness and capacity to adopt, implement, and scale genome-editing technologies.

Enumerators were identified, recruited, and trained through an online induction program on the use of ODK and associated data-collection tools. Training was conducted by the IT team and train-the-trainers (ToT) from the Consortium. All tools and platforms were pre-tested before deployment. Both methodological and data triangulation approaches were employed to reduce bias and enhance the validity and reliability of findings. All collected data were encrypted and stored on a secure, centralized server to ensure participant confidentiality and compliance with ethical standards.

2.7. Data Synthesis and Data Analysis

Data collected was synthesized and packaged into appropriate information, and, where applicable, tables were generated.

Variables investigated included, but were not limited to:

- i. signatory to multinational food and environmental agreements;
- ii. regulatory framework components;
- iii. applications received and approved (field testing, registration, and commercialization);
- iv. R&D institutions, personnel, and laboratory/field facilities;
- v. projects;
- vi. crops and livestock species; and
- vii. traits under GEd development.

2.8. Interactive Map

An interactive map, comparable to the Agenda 2063 Dashboard, was developed to visualize genome editing activities and capacities across Africa and is accessible at:

<https://genome.africa/>

3. RESULTS

3.1. Membership in Multilateral Environment Treaties and Agreements

Senegal is a signatory to many multilateral environmental and food agreements and treaties, including CAC, UNFCCC, CBD, the Cartagena Protocol, and the Nagoya-Kuala Lumpur Supplementary Protocol on Liability and Redress, Access and Benefit Sharing (ABS) (Table 1). Senegal signed the UNFCCC in 1992 and ratified it in 1994. It signed and ratified the UN CBD in 1998. It is also a party to the Nagoya-Kuala Lumpur Supplementary Protocol, which it signed in 2012 and ratified in 2016.

Table 1. Status of Country Participation in Key Multilateral Environmental Agreements (MEAs)

Multilateral Environmental Agreements (MEAs) / Treaties	Date of Ratification / Accession by the Country	Reference
Codex Alimentarius Commission	(1966). Member	https://www.fao.org/fao-who-codexalimentarius/about-codex/members/en/
Convention on Biological Diversity	1998	https://www.cbd.int/
Cartagena Protocol on Biosafety	2003 (ratified)	https://www.cbd.int/
UNFCCC convention on climate change	Signed 1992 Ratified 1994	National Determined Contribution on Climate Change for Senegal.
Nagoya Protocol on Access and Benefit Sharing (ABS)	Signed 2012 Ratified 2016	https://www.geneconvene.org/senegal-reg/?utm

3.2. Policies that Advance or Hinder Adoption of Modern Biotech and GEd

Senegal promotes modern biotechnology and genome editing through its updated biosafety legislation and alignment with continental frameworks. While most existing instruments were initially designed around GMOs and biosafety, the new law enables Senegal to meet its international commitments and maximize the benefits of modern biotechnology. A key policy that promotes the adoption of modern biotechnology, including GEd, in Senegal is:

National Biosafety Strategy (2025–2029): This is the regulatory blueprint designed to safely harness modern biotechnology in Senegal. Aligned with the country's national development frameworks, it translates the updated Biosafety Law of 2022 into practical actions that balance scientific innovation with strict environmental and health safeguards. Approval of this strategy in November 2024 marked a significant step toward protecting the environment, public health,

and food safety. The guidelines for this strategy align with a perspective on the responsible and safe development of modern biotechnology, including GEd. The strategy's direct relevance to modern biotechnology falls into a few key focus areas:

- ***Unlocking Agricultural & Health Biotech:*** It provides the operational guidelines to research, test, and deploy genetically modified organisms (GMOs) and modern biotech solutions aimed at combating crop diseases (like aflatoxin), pests, and vector-borne diseases like malaria.
- ***Aligning with Global Standards:*** The strategy ensures that Senegal's use of biotechnology complies with the Cartagena Protocol on Biosafety, managing risks through mandatory four-tier safety categorizations and strict risk assessments.
- ***Bridging Policy & Research:*** It resolves previous regulatory gaps, setting clear pathways for research institutions to innovate while ensuring bio-containment. This paves the way for safe integration of advanced genomics and bioinformatics into the local bioeconomy.
- ***Public Awareness & Participation:*** Building on prior legislative reforms, the 2025–2029 framework emphasizes transparent communication, public awareness, and mandated labelling to maintain public trust as biotechnology applications grow.

3.3. National Regulatory Framework

3.3.1. Biosafety Act/Law, Biosafety Regulations/Decrees, and GEd Guidelines

The Biosafety Law (No. 2022-20 of June 14, 2022) and the Regulation C/REG. govern the country's regulatory framework for GMOs and gene-edited products 04/09/2020 on GMOs and biotechnology products within the Economic Community of West African States (ECOWAS) and the Inter-State Committee for Drought Control in the Sahel (CILSS) (FAO, 2020). In addition, it has ratified several international protocols, including the CAC (see Table 2). The National Biosafety Authority (NBA), established under Law No. 2009-27 (NBA, 2009), is responsible for regulating the development, contained use, import, export, transit, transport, deliberate release into the environment, and placing on the market of genetically modified organisms (GMOs) and/or their derivatives.

The NBA operates under the Ministry of Environment and Ecological Transition (MEET). Through the National Biosafety Committee (NBC), the NBA conducts risk assessments for the import, export, handling, transit, confined use, release, or commercialization of GMOs and their derived products. The Authority is supported by two advisory bodies: the NBC and the Scientific and Technical Biosafety Committee (STBC). The NBC, composed of 32 members, addresses socioeconomic issues, while the STBC, comprising 17 members, addresses all scientific matters related to modern biotechnologies and their products. Following consultation

with these advisory bodies, the NBA submits its recommendations to MEET, the competent authority.

In Senegal, applications for confined field trials of GMO and modern biotech products are evaluated by the MEET through the Direction of Environment and Classified Establishments (DEEC) and the NBC. The process requires a comprehensive Environmental and Social Impact Assessment (ESIA) in compliance with the updated Biosafety Law (Law No. 2022-20). This initial step involves preparing a research proposal and dossier, followed by submitting the application to the Environment Ministry. A schematic illustration of how the biosafety application review process (risk assessment), approval, and post-release monitoring would be carried out in Senegal is presented in Figure 1.

Table 2. Regulatory and Institutional Landscape for Genome Editing in Senegal

Institutions	Mandate / Relevance to GEd	Regulatory instruments	Date of enactment or publication	Coverage/ scope	Reference
Ministry of Environment and Ecological Transition	Regulating the development, contained use, import, export, transit, transport, deliberate release into the environment, and placing on the market of genetically modified organisms or their derivatives.	Biosafety Law (No. 2022-20 of June 14, 2022)	2022	R&D, Commercialisation, Trade, etc.	https://biosecuritesn.sn/
		Regulation C/REG. 04/09/2020	2020	R&D, Commercialisation, Trade, etc.	https://www.fao.org/faolex/results/details/fr/c/LEX-FAOC206866/

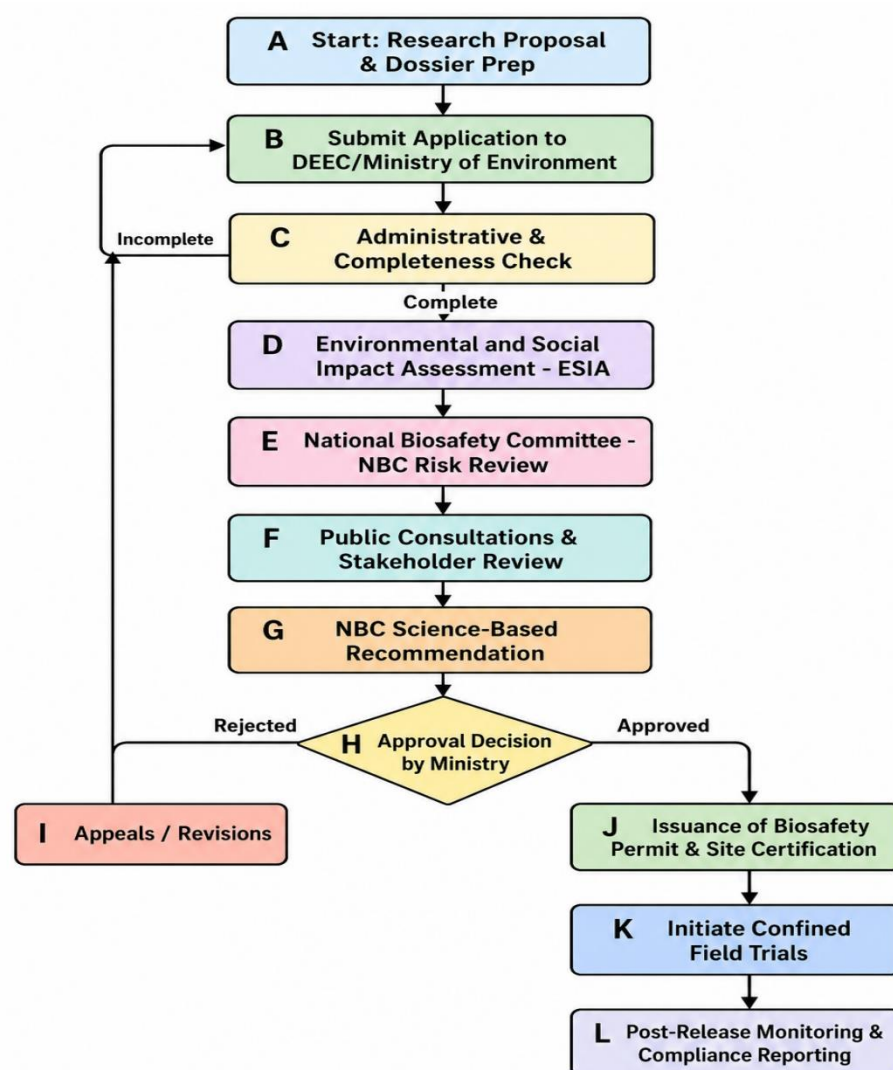


Figure 1. Biosafety application review and approval process in Senegal.

Source: National Biosafety Authority, 2022

3.4. Socio-economic considerations for decision-making

Socioeconomic issues are being addressed through the NBC's focus on food and nutritional security, the development of new crop varieties, the development of climate-resilient animal breeds, and biodiversity conservation.

3.5. Factors that Hinder or Advance Application of Modern Biotech and GEd

The application of modern biotechnology, including GEd, in Senegal is influenced by a complex interaction of political, geopolitical, social, human, cultural, and traditional factors. While Senegal has demonstrated growing interest in agricultural innovation, biosafety governance, and biotechnology research, several barriers continue to slow public acceptance,

policy implementation, commercialization, and investment in modern biotech and genome-editing technologies.

Political Factors: While Senegal passed the Revised National Biosafety Law in June 2022 to replace the restrictive 2009 legislation, operationalizing the new framework has been slow. The law's functionality depends on a set of implementing decrees (drafted by the National Biosafety Authority) that define specific risk assessments and categorization. Political deliberation and bureaucratic review of these decrees have dragged on, stifling clear pathways for researchers to apply genome editing techniques. Limited national research budgets are heavily influenced by political will, which must translate into direct national funding that is currently inadequate. Modern biotechnology and genome editing require immense investments in laboratory infrastructure, highly trained personnel, and intellectual property frameworks.

Geo-political Factors: Senegal relies heavily on agricultural exports to the European Union, and many EU markets maintain strict regulations due to consumer hesitancy regarding genetically modified food products. Thus, there is a fear that adopting modern biotech could threaten access to lucrative export markets if importing countries impose trade barriers or tariffs on genetically modified or gene-edited products. Additionally, Senegal's biosafety landscape must align with the broader Economic Community of West African States (ECOWAS) regional guidelines. While harmonizing policies across multiple nations ensures safe cross-border trade, it creates bureaucratic hurdles. Reconciling domestic priorities with regional ECOWAS policies slows down the rapid, localized approval processes needed for the quick deployment of genome-edited crops.

Social and Human Factors: There is widespread public scepticism, driven by limited literacy in modern biotechnology and genetics. Notably, there is limited awareness and understanding of modern biotechnology among the Senegalese public and various stakeholders. Concerns about "tampering with nature" and potential long-term ecological risks breed general technophobia. Resistance is often amplified by civil society campaigns that raise fears about the manipulation of traditional agricultural systems and the erosion of genetic diversity.

Culture and Traditions: Culturally, Senegal has a rich tradition of agroecology, and local agricultural practices rely heavily on organic fertilizers and community-led seed saving. Civil society networks strongly advocate for this traditional seed sovereignty, often framing modern biotechnology as an expensive distraction imposed by foreign entities. Smallholder farmers rely on domestic and regional (ECOWAS) markets that deeply value naturally grown produce. There is a deeply ingrained fear that adopting biotech crops could lead to rejection by local consumers and a loss of export opportunities.

3.6. Genome Editing Programs and Projects

Senegal currently has no genome-editing projects. However, basic training in molecular biology and genetics is available at the master's and doctoral levels. Research institutes and universities also offer training programs in fisheries and aquaculture, agroforestry, and livestock farming. Two scientists from Cheikh Anta Diop University and the Senegalese Institute for Agricultural Research (ISRA) were trained at the French Research Institute for Development (IRD) in Montpellier, France, to support GEd initiatives in Senegal. They are expected to launch a genome-editing project focused on some of the country's orphan crops.

3.7. Human Capital and Institutional Capacity

3.7.1 Summary of existing human capacity

Senegal possesses a rapidly growing human capacity in modern biotechnology and genome editing, primarily concentrated in agricultural research institutions such as the Institut Pasteur de Dakar (IPD) and Cheikh Anta Diop University (UCAD). While hands-on applications are scaling up, the nation is actively expanding its scientific workforce and regulatory structures. The Senegalese Institute for Agricultural Research (ISRA) is driving agricultural biotech. Local scientists and academic experts are participating in specialized training workshops and collaborating with organizations such as AUDA-NEPAD and UC Davis to adapt genome editing to enhance crop resilience and food security. For instance, on November 17 to 18, 2025, experts from Senegal's universities, the Senegalese Institute for Agricultural Research (ISRA), and the National Biosafety Agency (ANB), with support from AUDA-NEPAD, the University of California, Davis, IITA, and the USDA, met to discuss the potential of genome editing technology in Senegal's agricultural sector. All unanimously recognized the immense opportunities offered by this technology and the urgent need for Senegal to lay the necessary groundwork for its application in the country.

3.7.1.1. Women, Youth and Special Interest Groups involvement in Modern Biotech and GEd

Women, youth, and special interest groups have an increasingly significant role in advancing the need for Senegal to embrace modern biotechnology and GEd. Although the application of modern biotechnology and GEd has yet to be initiated in Senegal, these groups represent a major source of human capital, advocacy, and social acceptance. Senegal has seen gradual growth in women's participation in science, technology, engineering, and mathematics (STEM), including genomics, biotechnology, and bioinformatics. Female research scientists and lecturers are increasingly contributing to research and innovation ecosystems. Organizations such as JGEN Sénégal promote women's empowerment, leadership, entrepreneurship, and advocacy. Similarly, FESTIC (Association des Femmes Sénégalaises

des TIC) supports women's digital skills, innovation, entrepreneurship, and access to technology, which are important foundations for biotechnology awareness. Youth constitute a substantial proportion of Senegal's population and represent the country's greatest opportunity for building future biotechnology capacity. The creation of youth-led innovation hubs and entrepreneurship initiatives will open opportunities to integrate biotechnology with ICT, artificial intelligence, and data science. Special Interest Groups and civil society in Senegal shape the nation's landscape for the adoption of modern biotechnology, including GEd, through two competing fronts: Scientific advocacy for food sovereignty/health and agroecological grassroots movements that challenge industrial biotech. Entities such as the Senegalese Society of Human Genetics (S2GH) and various academic coalitions at Cheikh Anta Diop University (UCAD) function as primary drivers of scientific research. These groups work closely with international bodies like the African Biosafety Network of Expertise (ABNE) to align national policies with global research standards.

3.7.2. Research, Development and Academic Institutions

Senegal's biotechnology and genome editing landscape is transitioning from theoretical exploration to active development. Anchored by strong research and academic institutions, the sector benefits from a progressive biosafety framework. The Senegalese Institute for Agricultural Research (ISRA) is the primary public institution conducting agricultural R&D across plants, animals, forestry, and fisheries. Backed by the African Union Development Agency (AUDA-NEPAD), ISRA's focus includes developing climate-resilient crop varieties and ensuring seed sovereignty. International Consortia research institutions are heavily bolstered by partnerships with organizations such as the International Livestock Research Institute (ILRI), the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), and the International Institute of Tropical Agriculture (IITA).

3.7.3. Training and Professional Development

In 2022, 35 Senegalese scientists were trained in bioinformatics and genome sequencing by the Institute for Health Research, Epidemiological Surveillance, and Training (IRESSEF), part of the International Center for Research and Training in Applied Genomics and Health Surveillance (CIGASS) at Cheikh Anta Diop University (UCAD), Dakar (Table 3). AUDA-NEPAD and the NBA have also organized biosafety training workshops in Senegal, including a workshop titled "Strengthening the Capacity of Regulators, Monitoring, and Compliance," held in December 2021. These workshops ranged from two days to one (1) week. The training focused on bioinformatics, DNA sequencing, and biosafety communication. The AUDA-NEPAD GEd initiative has been crucial in providing financial support for some of these training

workshops, but additional funding is needed to increase participation from other public institutions.

Additionally, Senegal's higher education institutions, especially universities, are integrating plant and microbial biotechnology and population genetics into graduate programs (Table 4). These courses could potentially benefit from GEd technology.

Table 3. Overview of Training Programmes on Genome Editing

Institution / Organization	Training Programme	Target Audience / No. of Trainees per year	Frequency	Duration	Gaps Identified
Institut de Recherche en Santé de Surveillance Épidémiologique et de Formation (IRESSEF)	Bioinformatics and DNA sequencing	Students and Researchers	3 times/year	1 week/training	Limited participation from other public institutions. Need wider coverage. Also, a lack of funding (investment)
CIGASS à Université Cheikh Anta Diop (UCAD)	Bioinformatics and DNA Sequencing	Students and Researchers	4 times/year	1 week/training	Limited participation from other public institutions. Need wider coverage. Also, a lack of funding (investment)
AUDA-NEPAD GEd initiative	Biosafety training	Consumers, producers, students, researchers, and professionals	2 times/year	2 or 3 days/ training	Limited participation from other public institutions. Need wider coverage. Also, a lack of funding (investment)
National Biosafety Authority	Biosafety training, communication, and Biosafety Clearing House use	Consumers, producers, students, researchers, and professionals	10 times/year	2 days/ training	Limited participation from other public institutions. Need wider coverage. Also, a lack of funding (investment)

Table 4. Universities Offering Training on Biotechnology and GEd Related Courses in Senegal

Institution	BSc Program	Master's Program	PhD Program
Cheikh Anta Diop University	Fundamental life sciences Agro resources Plant & Microbial Biotechnology	Plant & Microbial Biotechnology Population Genetics Natural Products Biochemistry Plant phytopathology Agroforestry Botanic and Biodiversity	Plant & Microbial Biotechnology Population Genetics Plant phytopathology Agroforestry Botanic and Biodiversity
Gaston Berger University	Fundamental life sciences Agricultural and animal resources	Agronomic Sciences Aquaculture Sciences Food technology	Agronomic Sciences Aquaculture Sciences Food technology
Iba Der Thiam University	Fundamental life sciences Agricultural and animal resources	Agricultural science Animal production	Agricultural science Animal production
Cheikh Hamidou Kane Virtual University	Molecular Genetics and Bioinformatics	Molecular Genetics and Bioinformatics	N/A
Amadou Moutar Mbow University	Fundamental life sciences Agricultural and animal resources	Crop Production Animal production Food technology	N/A
Elhadj Ibrahima Niass University	Fundamental life sciences Agricultural and animal resources	Agronomic Sciences Animal production Food Technology	N/A
Assane Seck University	Fundamental life sciences Agroforestry	Agroforestry Agricultural Science	Agroforestry Agricultural Science
Alioune Diop University	Fundamental life sciences Agro-resources	Agricultural production	Agricultural production

3.7.3.1. Role of Ministry of Higher Education, Research and Innovation (MESRI) and the National Authority for Quality Assurance in Higher Education (ANAQ-Sup)

The recognition and quality of the degrees in biotechnology and GEd-related courses offered at the universities listed in Table 4 benefit from the roles of the **Ministry of Higher Education, Research and Innovation (MESRI)** and the **National Authority for Quality Assurance in Higher Education (ANAQ-Sup)**. MESRI handles the national policies and accreditation frameworks for all public and private higher education institutions. MESRI operates alongside a dedicated quality assurance and degree evaluation agency:

- **Ministry of Higher Education, Research and Innovation (MESRI):** The top-level government authority that determines the academic policies, approves new universities, and issues national degree accreditations.
- **National Authority for Quality Assurance in Higher Education (ANAQ-Sup):** An independent body operating under MESRI that evaluates training programs, accredits private universities, and sets the national quality and qualification standards for higher education degrees.

3.8. Infrastructure and Equipment

3.8.1. Universities

Most universities in Senegal support molecular research that could potentially integrate GEd into their research portfolios (Table 5). However, aside from the well-equipped laboratories at Cheikh Anta Diop University, universities are generally under-resourced and lack the specialized personnel needed to implement advanced agricultural technologies such as GEd.

Table 5. Status and Needs Assessment of Biosafety Laboratory Facilities by Universities

Institution	Type of Facility	Biosafety Level	Status	Limitations	Support Needed
Cheikh Anta Diop University	Nucleic acid sequencing	BSL 2	Fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable supply of power, maintenance challenge,	Specialized procurement/ waiver/ exemptions,
Gaston Berger University	No facility	No	not equipped	non-enabling national procurement law, inadequate funding, unstable supply of power, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Iba Der Thiam University	No facility	No	not equipped	non-enabling national procurement law, inadequate funding, unstable supply of power, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Alioune Diop University of Bambey	No facility	No	not equipped	non-enabling national procurement law, inadequate funding, unstable supply of power, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Assane Seck University of Ziguinchor	No facility	No	not equipped	non-enabling national procurement law, inadequate funding, unstable supply of power, and maintenance challenges	Specialized procurement/ waiver/ exemptions
El Hadj Ibrahima Niass University of Sine Saloum	No facility	No	not equipped	non-enabling national procurement law, inadequate funding, unstable supply of power, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Cheikh Hamidou Kane Virtual University	No facility	No	not equipped	non-enabling national procurement law, inadequate funding, unstable supply of power, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Amadou Moctar Mbow University	No facility	No	not equipped	non-enabling national procurement law, inadequate funding, unstable supply of power, and maintenance challenges	Specialized procurement/ waiver/ exemptions

3.8.2. Research institutions

Senegal has one central national agricultural research organization, the Senegalese Institute for Agricultural Research (ISRA) (*see Table 6*). However, its laboratory facilities are poorly equipped and underfunded, and lack specialized equipment such as high-throughput DNA sequencers. Bureaucratic procurement procedures often constrain them. Senegal also hosts Africa Rice, IRD and CIRAD, which are similarly ill-equipped for molecular research and would be limited in their capacity to leverage GEd technology fully should the country choose to adopt it.

Table 6. Status and Needs Assessment of Biosafety Laboratory Facilities by Research Institutions

Institution	Type of Facility	Biosafety Level	Status	Limitations	Support Needed
ISRA	Nucleic acid sequencing	BSL 2,	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable supply of power, maintenance challenge,	Specialized procurement/ waiver/ exemptions,
AfricaRice Centre	Not equipped	No	Not equipped	Lacks requisite facilities for biotech/GEd work	Specialized procurement/ waiver/ exemptions
IRD	Not equipped	No	Not equipped	Lacks requisite facilities for biotech/GEd work	Specialized procurement/ waiver/ exemptions
CIRAD	Not equipped	No	Not equipped	Lacks requisite facilities for biotech/GEd work	Specialized procurement/ waiver/ exemptions

3.8.2.1. Role of Ministry of Higher Education, Research and Innovation (MESRI) for Research in Senegal

The **Ministry of Higher Education, Research, and Innovation (MESRI)** is the primary governing body in Senegal responsible for regulating, coordinating, and promoting scientific research. MESRI supports and coordinates all research at ISRA and universities. While MESRI dictates overarching national policies and science and technology priorities, including modern biotechnology, research and innovation management is decentralized across several technical ministries and specialized agencies:

- **General Governance & Policy:** The MESRI oversees public higher education and defines the national strategic objectives for research.
- **Agriculture & Agribusiness:** Agricultural research is primarily coordinated by the Senegalese Institute for Agricultural Research (ISRA) under the Ministry of Agriculture.
- **Food & Technology:** The Institute of Food Technology (ITA) handles applied research in food processing under the Ministry of Industry.
- **Health & Bioethics:** Health research is managed by the Ministry of Health, working closely with the National Ethical Committee for Health Research (CNERS) and the Institut Pasteur de Dakar.
- **Ecological & Environmental:** The Ecological Monitoring Centre (CSE) coordinates environmental studies

3.9. Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/Breeds for Improvement Using GEd

In accordance with the Food Sovereignty Strategy (SSA) (2025–2034) and the Climate Smart Agriculture Investment Plan (CSAIP) (2025), alongside the broader Emerging Senegal Plan (PSE) (2014), several traditional and commercial agricultural crop value chains to drive both food security and economic growth were prioritized. For example, the primary cereals (grains) prioritized include rice, millet, and sorghum. Rice is specifically targeted to boost rural poverty reduction and cut imports. The crops with significant potential to benefit from the GEd interventions in Senegal are listed in Table 7. The GEd technology in trees poses additional challenges because of the difficulty of culturing woody plant tissues. However, GEd offers opportunities to enhance tolerance to both biotic and abiotic stresses in economically important species, whether for food production (e.g., fruit trees) or industrial uses such as timber and carpentry. The GEd technology can also help combat diseases, such as tsetse fly infestations and avian influenza, that threaten animal production. Additionally, the myostatin gene, considered an economically important genetic marker (Lee et al., 2024), can be targeted by GEd techniques to improve animal production (e.g., fish, poultry, and livestock).

Table 7. National Priority Crops that Could Potentially Benefit from GEd Technology

Organism / Species	Trait improvement of Interest	Socioeconomic Justification	GEd Potential (Low/Medium/High)	Existing R&D	Actual Annual Production Capacity (tons)	Expected Annual Production Capacity (tons)
Cowpea	Pest and Disease Resistance	Primary protein source and staple for rural households Used for human consumption, animal feed, and soil nitrogen-fixation. Supports smallholder incomes in the Sahel region.	High	No	65 000	130 000
Peanuts	Pest and Disease Resistance	Senegal is Africa's third-largest peanut producer. Oilseed supports household nutrition and provides a source of cash income. They are also a critical export commodity.	High	No	1 000 000	2 000 000
Tomatoes	Shelf-life increment	Widely consumed fresh and processed domestically; surplus exported to Europe. Supply supports local processing (paste and sauces) and underpins agribusiness growth.	High	No	45 000	90 000
Pepper	Pest and Disease Resistance	Important horticultural crop (e.g., pimento, hot pepper) in the coastal Niayes region. High value per hectare; both domestic consumption and export potential.	High	No	Data not available	Data not available

Rice	Bacterial wilt resistance, bacterial leaf streak	Staple food, with >2 Mt annual consumption; rice accounts for ~38% of cereal output (~1.5 Mt). High social and economic relevance.	High	No	1 346 000	3 000 000
Mango	Mango fruit fly resistance	A valuable source of income for farmers and contributes to the agricultural economy through agro-processing industries and export markets.	High	No	128 000	200 000
Chicken	Avian influenza and Newcastle disease resistance	Primary source of animal protein. Source of livelihood and income	High	No	114 200	200 000
Fisheries	High-feed conversion efficiency	Primary source of animal protein. Source of livelihoods and income	High	No	400 000	800 000
Millet	Striga resistant	Primary source of staple food Nutrition in rural zones	High	No	900 000 -1 000 000	

Source: Food Sovereignty Strategy - SSA (2025–2034); the Climate Smart Agriculture Investment Plan - CSAIP (2025) - Ouédraogo et al. 2026

3.10. Intellectual Property Rights and Benefit Sharing

Senegal's IP framework is shaped by its membership in the African Intellectual Property Organization and its adherence to international treaties, including the TRIPS Agreement. The country's IP laws aim to protect innovations, trademarks, copyrights, and industrial designs, creating a conducive environment for business growth and innovation.

Senegal has a legal framework for IP that aligns with both national and international standards. The key IP laws and regulations in Senegal include:

Intellectual Property Law – Law No. 2008–2009 of January 25, 2008, governs patents, trademarks, industrial designs, utility models, and trade secrets. This law outlines the procedures for registering and protecting industrial property rights in Senegal.

OAPI Uniform Law on Marks (Law No. 005/2002): This law indeed protects trademarks across all OAPI member states, including Senegal. The law is part of the Bangui Agreement, which standardizes intellectual property regulations across its member states.

The Copyright Law (Law No. 2008–2009 of January 25, 2008): Covers copyrights and related rights. It protects literary and artistic works, including music, film, software, and databases. Senegal is a signatory to the Berne Convention, which further strengthens its copyright regime.

Law on the Protection of New Plant Varieties: Senegal is a member of the International Union for the Protection of New Varieties of Plants (UPOV) and has legislation in place, Law No. 9-94 on the Protection of Plant Varieties, that protects plant breeders' rights. This law encourages the development of new plant varieties and ensures that breeders can benefit from their innovations. Senegal is compliant with regional agreements such as **OAPI (Bangui Agreement, 2015)**, which governs IPRs in 17 African countries. A summary of these IPRs is presented in Table 8.

Table 8. A summary of Intellectual Property Rights (IPRs), their Institutions and Laws in Senegal

Intellectual Property Rights (IPRs)	Organization / Institution	Laws (Acts) / Regulations / Protocols / Treaties
Patents, Trademarks, Industrial Designs, Geographical Indications	Senegalese Agency for Industrial Property and Technological Innovation (ASPIT).	Legal framework of OAPI, i.e., Bangui Agreement (14 December 2015 Act)
Copyright and Neighboring Rights	Copyright of Office of Senegal (SODAV)	Law No. 2008-09
Plant Breeders Rights		Law No. 9-94

Patents, Trademarks, Industrial Designs	African Intellectual Property Organization (OAPI)	Bangui Agreement (14 December 2015 Act)
International patent applications	World Intellectual Property Organization (WIPO)	Patent Cooperation Treaty (PCT) 1970 (amended 1979, 1984, 2001)
International registration of marks		Madrid Protocol 1989, enforced 1995
Industrial Designs		Hague Agreement (Act 1960); Geneva Act 1999

3.11. Private Sector Participation

Senegal's private sector involvement in genome editing and modern biotechnology is still evolving, with most efforts currently led by public institutions and international partnerships. However, local private sector participation is beginning to emerge in specific areas (*see Table 9*). Many GEd initiatives in West Africa are implemented through international consortia that involve African research centers and CGIAR partners, with Senegalese laboratories and private contractors serving as implementation partners. With NBA authorization, the private sector is expected to spearhead the application of GEd in breeding programs to improve production quality and quantity. The private seed companies include Tembédou et Fils, SEDAB, Tropicasem, Bayer, and Syngenta. These companies play a crucial role in supplying quality seed and supporting smallholder farmers. Bayer and Syngenta, in particular, play a significant role in communicating and advocating for agricultural biotechnology.

Table 9. Overview of private sector companies involved in seed production & distribution, crop Improvement & biotech/GEEd activities in Senegal

Company	Type (Multinational / Regional / National	Biotech/ GEEd Activities	Partners	Challenges Faced	Investment Interest
ETB Tambédou et Fils	National	No GEEd	ISRA/CNRA, RESOPP, Technisem, Agrosemens, SSG	State dominance and subsidies, complex quality certification, weak regulation of the seed market, counterfeits	Seed multiplication and distribution
SEDAB	National	No GEEd	ISRA, myAgro, AfricaRice, RESOPP, MoA, FAO, GIZ	State dominance and subsidies, complex quality certification, weak regulation of the seed market, counterfeits	Seed breeding and R&D, seed production, processing & distribution
Tropicasem	National	No GEEd	Technisem, Novalliance Group, COOPROFEL, Cordaid, University of Thies, UCAD	State dominance and subsidies, complex quality certification, weak regulation of the seed market, counterfeits	Seed breeding and R&D, seed production and distribution
Bayer	Multinational	No GEEd	Solevo Group, ISRA, UNIS, myAgro, IITA	State dominance and subsidies, complex quality certification, weak regulation of the seed market, counterfeits	Seed breeding and R&D, seed production and distribution
Syngenta	Multinational	No GEEd	Sustainable Agriculture Foundation (SAF), Tropicasem, SEDAB, Technisem, AfricaRice, AATF, ISRA, UNIS,	State dominance and subsidies, complex quality certification, weak regulation of the seed market, counterfeits	Seed breeding and R&D, capacity building, small scale seed production; Seeds2B initiative.

Private-sector presence in Senegal presents significant opportunities for future engagement in modern biotechnology and GEd technologies once the biosafety law is operational. The sector, like in other African countries, faces several challenges. On the other hand, these private-sector players will have opportunities when Senegal operationalizes its biosafety law and adopts modern biotechnology and GEd. The current challenges and future opportunities associated with biotech/GEd adoption are presented in Table 10.

Table 10. Challenges and Opportunities of Engaging in the Private Sectors

Challenges	Opportunities
State dominance and subsidies	Senegal can position itself as a regional biotechnology hub by developing clearer application and approval pathways.
Complex quality certification;	The financing gap creates opportunities for public-private partnerships (PPPs), impact investors, agritech incubators, and regional innovation funds.
Weak regulation of the seed market and limited access to rural distribution points.	Shared innovation platforms, regional centers of excellence, and private laboratory service providers can emerge to supply biotechnology services.
Seed counterfeits	The skills gap creates opportunities for universities, private training institutions, and international firms to establish biotechnology training programs.
Weak commercialization pathways from research to industry	Multi-stakeholder innovation platforms can improve collaboration and accelerate commercialization.
Climate change, emerging pests, and diseases increase agricultural risks	There is strong market demand for improved crops with drought tolerance, pest and disease resistance, salinity tolerance, and enhanced nutritional traits.

3.12. Funding and Investment Landscape

Funding for biotechnology-related agricultural research comes from various sources, including international organizations and the government (*see Table 11*). Key players in this area include the NBA and AUDA-NEPAD, which promote public trust in agricultural research and biotechnology. AUDA-NEPAD supports initiatives to implement guidelines for modern biotechnology and communication across Africa, including genome editing, and collaborates with the African Agricultural Technology Foundation (AATF) to promote locally tailored, science-driven genome editing innovations. The failure to implement the relevant decrees contributed to a lack of funding for research on GEd and GMOs.

Table 11. Overview of National and other Potential Funding Sources for Genome Editing

Funder/Donor	Organization Type	GE ^d Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
Government	None	None	None	None	None	None
AUDA-NEPAD	Development partner	None	None	None	None	None

4. RECOMMENDATIONS AND POLICY OPTIONS

To successfully implement genome editing in Senegal, it is essential to prioritize several key areas: establishing a robust regulatory framework, attracting investment, strengthening institutional capacity, promoting collaboration, and advancing effective scientific communication and advocacy.

4.1. Recommendations

Regulatory Framework: The implementation of Senegal’s Biosafety Law depends on the publication of the relevant implementing decrees, which are required for the use and commercialization of GE^d in both confined and open environments.

Capacity Building: Universities and research institutes require targeted training programs in genome editing, bioinformatics, molecular biology, tissue culture, risk assessment, bioethics, and intellectual property management. By establishing modern laboratories equipped with appropriate equipment, alongside mentorship programs and exchange opportunities, Senegal can develop a skilled pool of GE^d scientists and regulators capable of both innovating and governing effectively.

Investment Strategies: Investment strategies should not rely solely on traditional donor funding. To support GE^d research from early-stage discovery through product development, Senegal should establish innovative financing mechanisms, including public-private partnerships, competitive grants, and national research endowment funds. Public investment can be complemented by tax incentives and venture capital for agri-biotech startups, strengthening the overall research environment. Incorporating genome editing into national agricultural investment plans can align research priorities with food security, climate resilience, and industrial crop development.

Collaborations and Partnerships: Collaboration and partnerships are vital to amplifying impact. Senegal should deepen its involvement in regional initiatives under AUDA-NEPAD,

ECOWAS, and ARIPO to share expertise, harmonize regulatory frameworks, and access joint funding programs. Within the country, stronger research networks among research institutes, universities, farmer-based organizations, and the private sector will enhance the translation of innovations into practical solutions. Structured partnerships with international research bodies, such as the CGIAR, and advanced laboratories abroad would also provide access to technical expertise and research equipment that are often difficult to source locally.

Communication and Advocacy: Effective communication and advocacy are essential to societal acceptance of modern biotechnologies. The country needs to invest in dialogic communication and advocacy to demystify technologies such as GEd, build public trust, and actively involve communities in decision-making. Dialogue platforms that explain the benefits, risks, and regulatory safeguards of genome editing in an accessible language should include civil society, farmer associations, policymakers, traditional communicators, and authorities. Tailored media campaigns and school-level biotechnology education can help cultivate a more informed society prepared to embrace innovation while holding institutions accountable.

4.2. Policy Options

To harness the full potential of genome editing and advance national priorities in food security, economic growth, and sustainable development, numerous policy options are outlined in Table 12.

Table 12. Policy matrix linking Issues → Gaps → Recommended actions → Lead Institutions

Issues	Gaps	Recommended Actions	Lead Institutions
Regulatory framework for genome editing	Biosafety Law (2022) and regulations focus on GMOs, not genome editing; there are no clear guidelines.	Publish the implementing decrees; adopt a tiered risk-based assessment; align with AU, ECOWAS, and international standards.	National Biosafety Authority (NBA), Ministry of Environment, Ministry of Higher Education, Research, and Innovation, Parliament.
Institutional and human capacity	Limited expertise in genome editing, biosafety risk assessment, and bioethics; inadequate laboratories.	Establish modern biotechnology laboratories; build technical and regulatory capacity through training, mentorship, and international exchanges.	Universities, Research Institutes
Funding and investment in research	Over-reliance on donor support, limited public investment, and no dedicated biotech funding streams.	Create a national biotech/genome-editing fund; incentivize private-sector participation; establish public-private partnerships.	Ministry of Finance, Ministry of Higher Education, Research, and Innovation, Ministry of Agriculture, Private Sector
Access to research tools and materials	Delays and high costs in importing reagents and equipment; weak supply chains.	Streamline customs processes for scientific imports; build regional biotech hubs; promote local production of key reagents.	Ministry of Finance, Research Institutes, Universities
Research collaboration and networks	Weak coordination among local institutions; limited participation in regional/global biotech initiatives.	Strengthen national biotech networks; join and leverage AUDA-NEPAD, ECOWAS, African Regional Intellectual Property Organization (ARIPO), and CGIAR platforms for knowledge and resource sharing.	Universities, Research Institutes, Ministry of Higher Education, Research, and Innovation.
Public awareness and communication	Low awareness and understanding; mistrust from GMO debates; weak	Develop inclusive science communication strategies and leverage traditional leaders, farmer groups, media, and schools to build trust.	Farmer-Based Organizations (FBOs) and the National Biosafety Authority.

Issues	Gaps	Recommended Actions	Lead Institutions
	engagement with farmers and communities.		
Socio-cultural acceptance	Religious and cultural skepticism; limited integration of local knowledge into biotech policies.	Institutionalize consultations with traditional authorities, faith-based groups, and communities in biotech governance.	Ministry of Higher Education, Research, and Innovation, Ministry of Culture and Religious Affairs, Ministry of Communication.
Intellectual property rights (IPR)	Weak IPR enforcement in biotechnology; little experience handling patents for GMOs/genome editing.	Build IPR capacity; strengthen alignment with ARIPO and WIPO frameworks.	Registrar-General's Department, ARIPO Focal Point, Ministry of Justice & Attorney General's Department.
Regional and international obligations	Slow domestication of the Nagoya Protocol and weak enforcement of Access and Benefit Sharing (ABS).	Fully implement Nagoya and Kuala Lumpur Protocols; strengthen ABS laws; ensure equitable benefit sharing of genetic resources.	Environmental Protection Agency (EPA), Ministry of Higher Education, Research, and Innovation, Attorney General's Department.

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ANNEXES

Annex 1: List of Institutions and Resource Persons Involved in the Interview

SN	Name	Sector	Ministry/Department/Institution/ Organization	Position	Contact Details
1	Aliou Ndiaye	Regulatory	National Biosafety Authority (NBA)	Executive Director	aliou.ndiaye@ucad.edu.sn
2	Ndjido Ardo kane	Research Institutions	ISRA	Research scientist	ndjidokane@gmail.com
3	Romaric Nguepjop	Research Institutions	CIRAD	Research scientist	joel-romaric.nguepjop@icard.fr
4	Mamadou Sock	Research Institutions	AfricaRice	Research engineer	sockm8427@gmail.com
5	Malick Diouara	Universities	Cheikh Anta Diop	Associate Professor	malick.diouara@esp.sn
6	François A. Badiane	Universities	Cheikh Anta Diop	Assistant Professor	francois.badiane@ucad.edu.sn
7	Gora Diop	Universities	Cheikh Anta Diop	Associate Professor	gora.diop@ucad.edu.sn
8	Arfang Diamanka	Universities	Cheikh Anta Diop	Assistant Professor	arfang.diamanka@ucad.edu.sn
9	Anicet Manga	Universities	Gaston Berger University	Associate Professor	anicet.manga@ugb.edu.sn
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